

AI-ASSISTED TAX RETURN REVIEW

# CPA Trace

*Every number, back to its source.*

July 2026

# The AI is fast But not trustworthy

- Tax firms now use AI to read client documents and auto-fill returns. **Extraction is fast.**
- The CPA who signs the return is **personally liable** for every number. An AI that is 95% right is still wrong somewhere.
- **If re-checking the AI takes as long as doing the work, the AI is worthless.**

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*The bottleneck isn't extraction. It's trust.*

# Make verifying the AI faster than redoing the work

01

**Allocate attention in proportion to risk.**

High-confidence figures: confirm in one click. Uncertain ones: stop the reviewer and show the evidence.

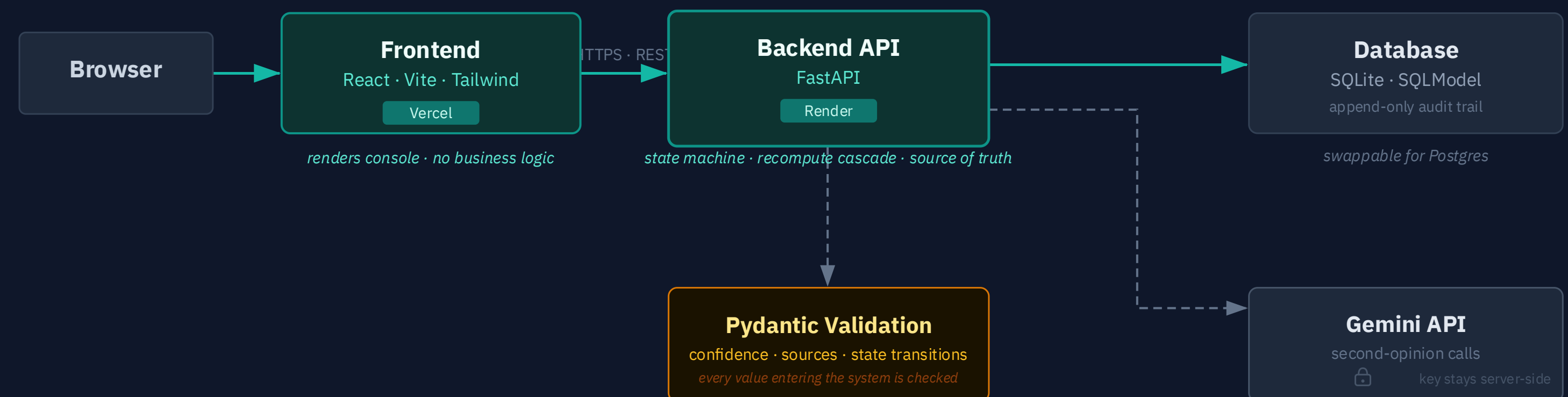
02

**The AI advises. The human decides.**

The system never silently changes a number a CPA will sign. Every value that lands is a human decision, recorded.

*Speed on the safe fields, scrutiny on the risky ones.*

# Web-app architecture



*Frontend renders. Backend decides and remembers. Validation guarantees the data is always well-formed.*

# Field Data Model

| FIELD — the core data structure |                                                                       |                                                |
|---------------------------------|-----------------------------------------------------------------------|------------------------------------------------|
| value                           | 142,300                                                               | the number                                     |
| provenance                      | extracted · computed · carried-forward · client · conflicted          | WHERE it came from (5 types)                   |
| state                           | ai_suggested · needs_review · verified · corrected · pending · locked | WHERE in review (6 states)                     |
| confidence                      | 0.0 – 1.0, or absent                                                  | how sure the AI was (only when a model judged) |
| edit_history                    | [ {who, when, old→new, why}, ... ]                                    | every change, ever — append-only               |

## INDEPENDENT AXES

**Provenance** — where did this value come from?

**State** — where is it in the review process?

A field has exactly one of each, in any combination.

Build this object once and every feature is one view of it — traceability renders provenance, affordances render state, the AI layer renders confidence and history.

*One well-designed object, rendered many ways.*

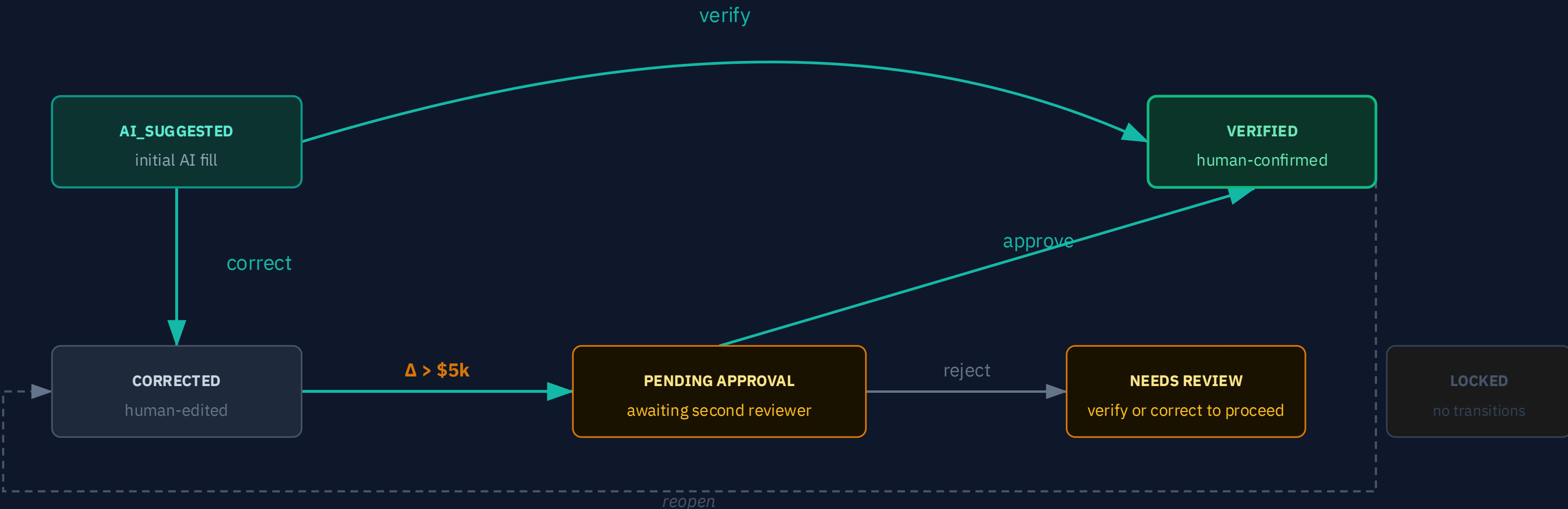


# Not every number comes from the same place

| TYPE                     | WHERE THE VALUE CAME FROM   | EVIDENCE SHOWN                                       |
|--------------------------|-----------------------------|------------------------------------------------------|
| ● <b>Extracted</b>       | Read off a document         | The page, exact region highlighted                   |
| ● <b>Computed</b>        | A formula over other fields | Formula + clickable inputs (no source page)          |
| ● <b>Carried-forward</b> | Last year's filed return    | The prior return                                     |
| ● <b>Client-provided</b> | A questionnaire answer      | Question + who answered — <b>no confidence score</b> |
| ● <b>Conflicted</b>      | Two documents disagree      | Both sources + AI recommendation — human resolves    |

*Traceability isn't one pattern — it's five, each handled honestly.*

# Six states, and only legal moves between them



**Sign-off rule:** a return cannot be filed while any field is unreviewed.

**Materiality routing:** corrections > \$5,000 auto-route to a second reviewer.

**Enforcement:** the backend rejects illegal transitions — the machine cannot reach an invalid state.

*Review isn't a checkbox — it's an enforced workflow.*

# One correction, all the way through the stack

- 1 Reviewer edits a field in the browser  
React · no business logic held client-side
  - 2 POST /api/fields/{id}/correct  
payload: { value, note }
- 
- SERVER-SIDE
- 3 Pydantic validates the payload  
rejects if malformed — type-checked before anything runs
  - 4 State machine applies the change  
→ corrected, or → pending\_approval if  $\Delta > \$5k$
  - 5 Cascade recomputes dependent totals  
interest total re-sums; its prior sign-off is voided
  - 6 Audit trail appended  
who, when, old → new, why — append-only, never overwritten
- 7 Updated fields returned as JSON
  - 8 UI refreshes  
corrected field and every recomputed total update

## STEPS 3–6: SERVER-SIDE

The frontend holds no business logic. Validation, state, cascade, and recording all happen in the backend.

## STEP 6: AUDIT TRAIL

Append-only — nothing is ever overwritten or deleted. Every correction is permanent history.

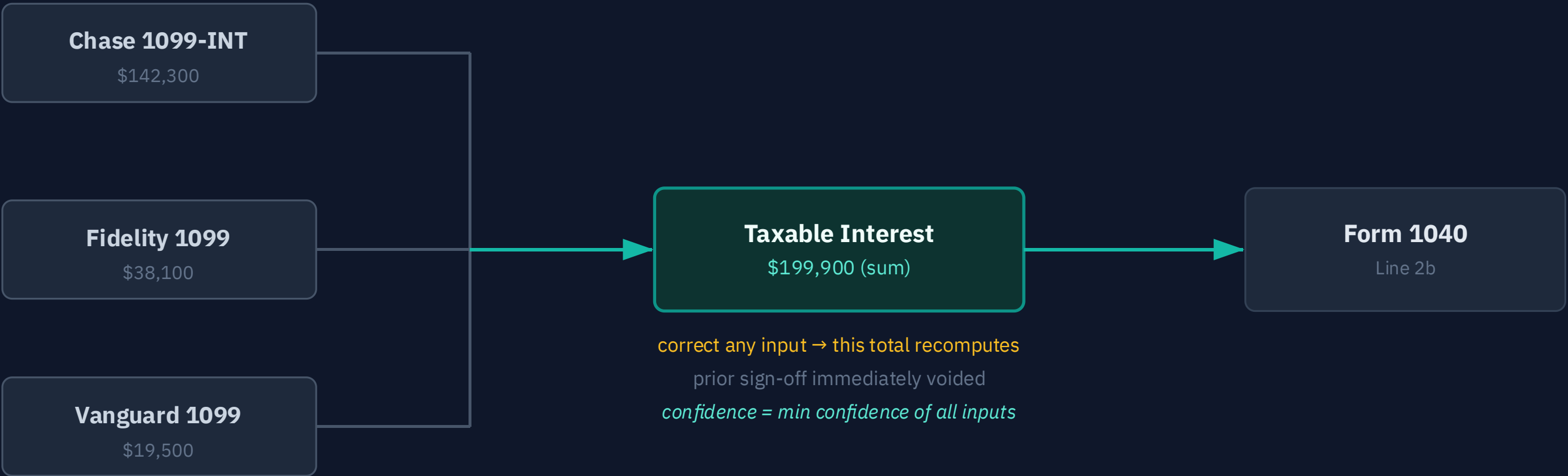
## STEP 5: CASCADE

A corrected input and a stale total can't coexist. The cascade enforces internal consistency.

*Frontend renders; the backend validates, decides, recomputes, and remembers.*



# Change one input, and everything that depends on it updates



*The system won't let a corrected input and a stale total coexist.*

# Six design tensions of a tax platform — and the idea behind each

## TRUST & PROVENANCE

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### Source Traceability

A CPA can't trust a figure they can't trace.

→ *Every field carries its origin. Click a number: source opens with the region highlighted.*

### Clickable vs. Editable

AI, verified, locked, and client values sit side by side.

→ *Six-state visual grammar. Only needs-review is styled loud.*

### Trustworthy AI

Showing every detail overwhelms; showing none hides risk.

→ *Confidence as guidance. Second opinion optional. Correction in one click, always recorded.*

## WORKFLOW

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### Actionable Dashboard

A reviewer with hundreds of returns needs direction, not a spreadsheet.

→ *Queue ranked by deadline, flags, and blockers. Waiting ≠ actionable.*

### Return Status & Progress

"In progress" means different things to clients and staff.

→ *One status, two renderings: reviewer detail + client plain language.*

## COLLABORATION

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### Client & CPA Collaboration

Questions scatter across email, detached from the documents they concern.

→ *Discussions live on the specific field they concern. Internal vs. client notes clearly separated. Not another inbox.*

# The model is stubbed by design, The trust layer is the real work

## Confidence Policy

Thresholds route low-confidence reads to review. Scores are never shown without a reason attached.

## Suppressed Confidence

No score where no model judged — client answers, prior-year carryforwards. A fake score would devalue every real one.

## Audit Trail as Evaluation Set

Every correction is a labeled example. Corrections clustered at low confidence: model is calibrated. At high confidence: the alarm that it isn't.

## Confidence Inheritance

A computed total inherits the *minimum* input confidence — one weak read poisons the result. Not an average, which would hide it.

## Autonomy Boundaries

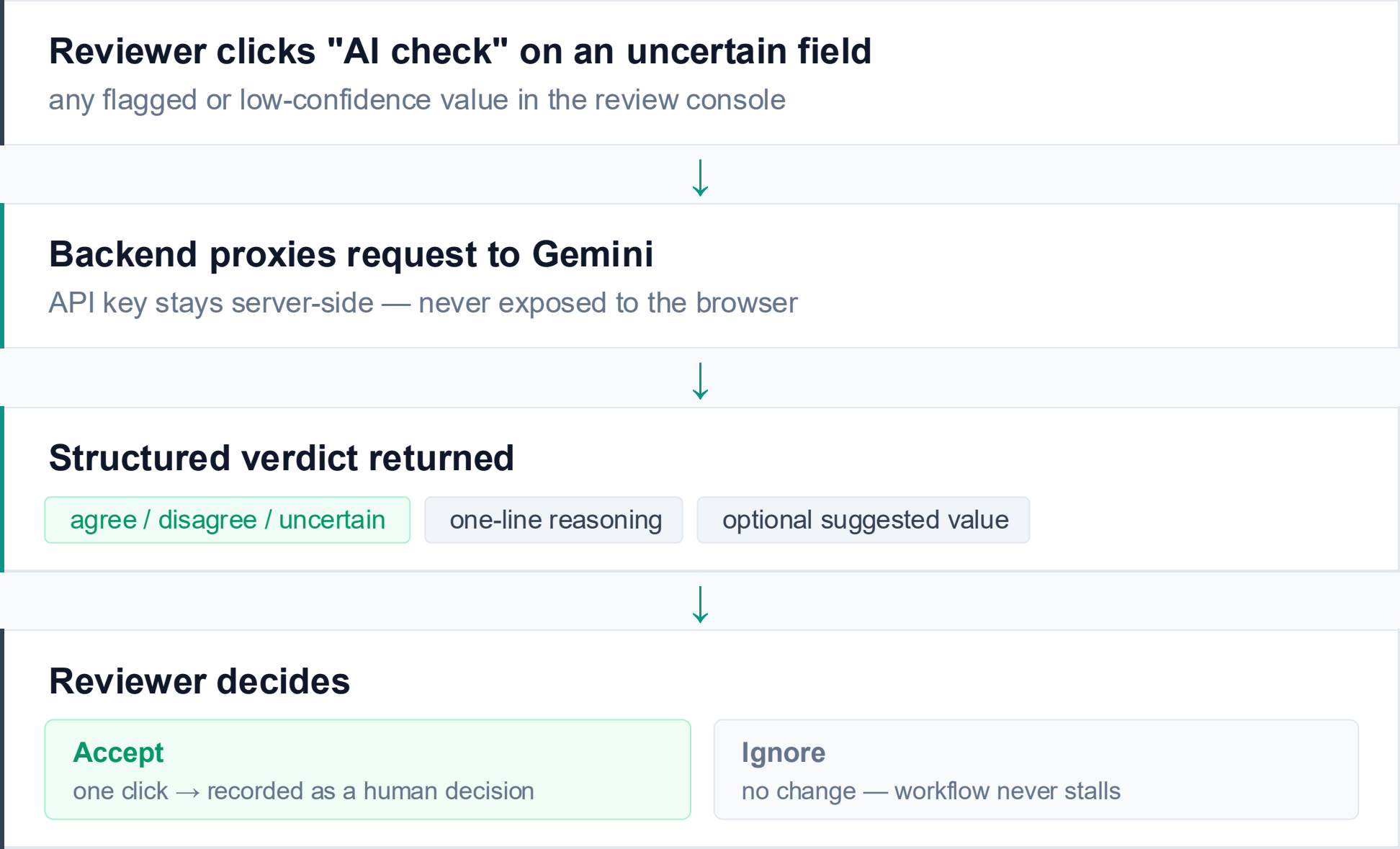
On a document conflict, the AI recommends and explains its reasoning. It never resolves. A human does.

## No Ground Truth by Nature

Errors surface only when a liable human disagrees. The human's decision is the label, generated one review at a time.

*Trustworthy AI is an interface and workflow problem as much as a model problem.*

# A second model, as an advisor — never an autopilot



## SECURITY

The API key is held server-side. Every model call is proxied through the backend — the frontend never sees it.

## AUTONOMY BOUNDARY

The second opinion is advisory. It can suggest; it never changes a value on its own. Accepting it is a human click, logged permanently.

## GRACEFUL FALLBACK

If the model is unavailable, a sensible fallback is returned. The review workflow never stalls on a model failure.

*The AI proposes. The human disposes if not reliable.*

# Honest about what's real and simulated

- SIMULATED (BY DESIGN)

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## Document extraction

No OCR. Source docs are mocks; extracted values, confidence, and flags are authored. Swapping in a real model changes nothing downstream - it produces the same validated shape the interface already consumes.

## Upload flow

Stands in for an extraction pass.

## Authentication

Guest login as a fixed reviewer.

- REAL

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- Review state machine - verify / correct / resolve / approve / reject, legal transitions enforced
- Recompute cascade - correct an input, dependent totals recompute server-side and lose prior sign-off
- Sign-off rule - can't file while any field is unreviewed
- Second-reviewer routing for material corrections
- Append-only audit trail
- Live second-opinion model call, with graceful fallback
- Dashboard ranking and derived status logic

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*The model is stubbed because the hard part isn't the extractor — it's the trust layer that lets a liable professional act on AI output in seconds. That layer is the product.*